Roadmap for AI project

Timeline & De-risking factors

Executive summary

Problem: Up to 80% of AI initiatives fail.

Reality: In 2025, 42% of companies scrapped their AI projects due to rising costs, poor data quality, or lack of clear business value.

Solution: Apply methodology eg PMI. Turn AI's unpredictability into repeatable, manageable processes.

Estimated timeline for MVP delivery

AI patterns	1: Biz udrsd	2: Data udrsd	3: Data prep	4: Dev	5: Eval	6: MLops
Primary De-risking Goal	**Justify ROI & Pattern**	**Technical Feasibility**	**Set Performance Ceiling**	**Build vs. Buy Decision**	**Real-world Stress Test**	**Prevent Model Drift**
Recognition	1 Week	2 Weeks	6-8 Weeks	3 Weeks	2 Weeks	Ongoing
GenAI	1 Week	1 Week	2 Weeks	2 Weeks	1 Week	Ongoing
Autonomous Systems	2 Weeks	2 Weeks	12+ Weeks	6+ Weeks	4+ Weeks	Ongoing
Hyper-Personalization	1 Week	2 Weeks	6-10 Weeks	3 Weeks	3 Weeks	Ongoing
Predictive Analytics	1 Week	2 Weeks	4-6 Weeks	3 Weeks	2 Weeks	Ongoing
Patterns & Anomalies	1 Week	2 Weeks	4-6 Weeks	3 Weeks	4-6 Weeks	Ongoing
Conversational	1 Week	2 Weeks	2-4 Weeks	2 Weeks	2 Weeks	Ongoing

Phase 0: 7 patterns of AI

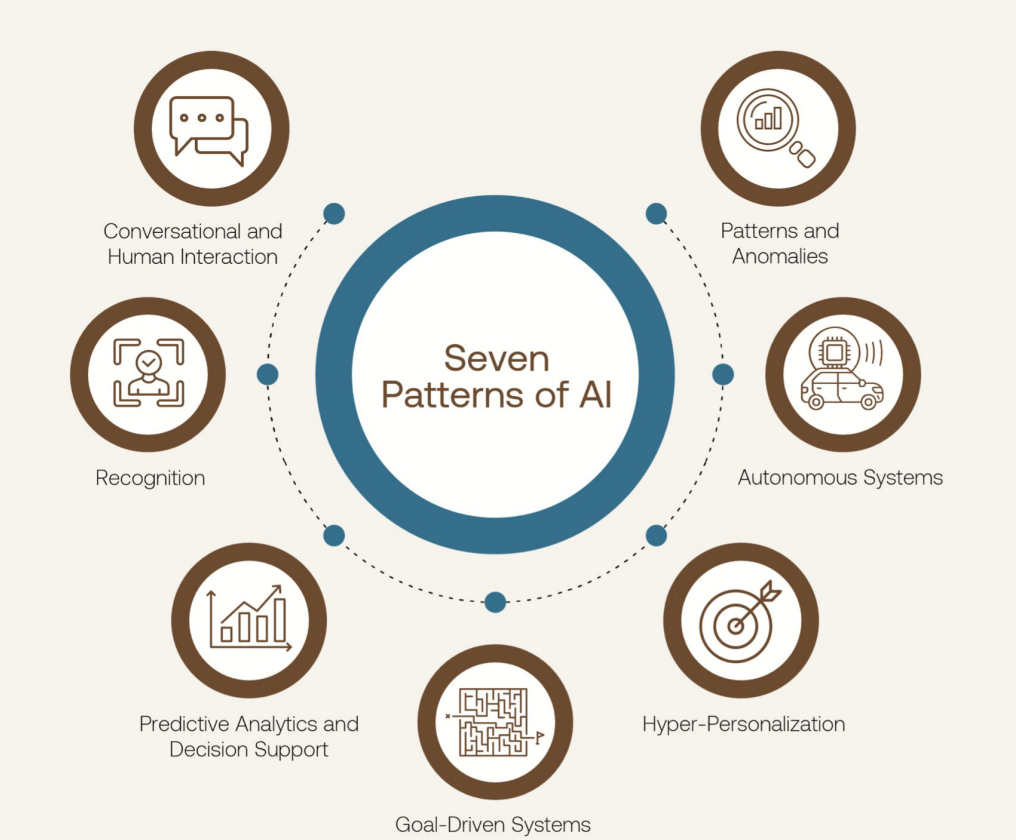

Figure 1. The Seven Patterns of AI

Source: PMI

Categorising.

Identify the specific pattern of the project.

Determine technical needs and risks early.

For project owners

Clarifies resource allocation & critical "make-or-buy" decisions.

For AI engineers

Defines the technical requirements.

Eg the need for computer vision experts vs linguists.

De-risk strategy

Starting with a single pattern
Reduce complexity
Speed up delivery.

Phase 1: Business understanding

Evaluate whether a problem actually requires AI.

For project owners

Prevents implementing for the sake of using the AI rather than addressing a specific problem.

Define SMART ROI goals and a MVP to deliver value in weeks.

For AI engineers

Identifying the AI pattern early.

Set the technical strategy before code is written.

De-risk strategy

Traffic Light Check: Green light requires clear, measurable success criteria agreed upon by all stakeholders.

Phase 2: Data understanding

Assess to ensure the technical feasibility of the data.

quantity / format / real time data or in batches / reliability and consistency.

For project owners

The primary Go/No-Go checkpoint.

If data is poor or inaccessible, stop or pivot to protect the budget.

For AI engineers

Perform systematic bias checks early. No algorithm can fix a foundationally biased or poor-quality dataset.

Phase 3: Data prep

Transform raw data into structured, AI-ready datasets.

80% of weight overall.

For project owners

Largest investment

Sets the performance ceiling for the entire project

For AI engineers

Build automated pipelines to enable repeatable iterations & easier long-term maintenance.

Phase 4: Model dev

Create functioning AI systems through custom builds, off-the-shelf tools, or fine-tuning foundation models.

For project owners

"Build-vs-Buy-vs-Integrate" decision.

Custom models create proprietary IP, while off-the-shelf tools offer speed but potential vendor dependency.

For AI engineers

Implement explainability tools (like SHAP or LIME) to ensure the model's logic is understandable and not a "black box".

Phase 5: Model eval

Testing under real-world conditions.

Verify the model delivers the business value defined in Phase 1.

For project owners

Deployment only happens after proof of measurable business process improvements (eg. 30% time savings).

For AI engineers

Conduct adversarial robustness testing to ensure the model remains stable under stress.

Phase 6: Model ops

Transition to live production with established MLOps for continuous performance assurance.

For project owners

Deployment is the beginning, not the end.

Establish ongoing monitoring to protect your investment from model and data drift.

For AI engineers

Establish versioning and retraining pipelines to handle shifting data patterns automatically.

Strategic de-risking

Phase 1 Go/No-Go feasibility checks.

Phase 2 Prioritise data feasibility.

Protect the budget by confirming the organization has the right "raw material" before the engineer begins development

Pattern Mapping: Use the 7 Patterns of AI to identify specific risks early.

eg "Recognition" requires longer timelines for labelling; "Autonomous Systems" carries higher liability risks.

Operational de-risking

Reputational Protection: layers of trustworthy AI into every phase.

De-risking Legal Liability: Privacy and regulatory compliance (eg GDPR) as core requirements.Prevent future legal liabilities and reputational damage.

Accountability: Maintain audit trails and clear decision-making authority matrices. Ensure human responsibility for AI-driven outcomes.

Technical de-risking

Commit to serious eval. Ensure the model works in real-world business settings.

MLOps infra to detect model and data drift.

Continuity Planning: Build contingency plans for technology shifts or vendor failures. Ensuring the AI system remains a sustainable asset.

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